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Research article

Rethinking radiology reports: A survey of referring physicians' perspectives

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ABSTRACT

Background: Radiology reports play a crucial role in clinical decision-making. High report quality and completeness are essential for efficient patient care. However, the clarity and comprehensiveness of radiology reports are often points of contention among referring physicians. This study evaluated referring physicians' views on the quality and clinical utility of radiology reports.

Methods: A prospective, anonymous online survey was conducted from June 2023 to March 2025, targeting 258 practicing physicians in Germany, including 93 internists, 90 surgeons and 75 general practitioners. The survey included rating scales, multiple-choice questions and net promoter scores (NPS).

Results: Referring physicians' satisfaction with the completeness of radiology reports was moderate, with an average score of 38.4 ± 24.3 on a scale from −100 to + 100. Among the specialties surveyed, surgeons reported the highest level of dissatisfaction ($p < 0.05$). Internists and surgeons showed significantly stronger preferences for structured reporting compared to general practitioners ($p < 0.05$). According to the surveyed referring physicians, the most frequently missing information in reports included "determination of a diagnosis" (39 %) and "recommendations for further diagnostic measures or follow-up examinations" (23 %). A large majority of referring physicians (84.9 %) found multidisciplinary case conferences to be valuable (5–7 out of 7 stars) for enhancing their understanding of reports.

Conclusion: Reporting preferences vary across specialties and radiologists must address the clinical needs of referring physicians, who are the primary audience for radiology reports. Integrating imaging into multidisciplinary meetings can further improve radiology report comprehension of referring physicians.

1. Introduction

Radiology reports are pivotal in bridging diagnostic imaging with clinical decision-making, serving as a critical communication tool between radiologists and referring physicians [1]. Although some referring physicians can analyze imaging studies themselves, a report from a radiologist adds significant value by offering a more detailed and accurate assessment [2–4].

Radiology reports traditionally rely on free-text reporting using dictation software, offering flexibility but varying in formats, completeness, and transcription accuracy. Reporting styles often reflect individual preferences of radiologists. While some tend to write detailed and extensive reports, others focus on concise, question-oriented descriptions. Attributes such as clarity, completeness, structure, and relevance define high-quality reports. However, a lack of standardization may impact their consistency and interpretability [5,6].

Contrary to free-text reporting, structured reporting offers an alternative approach. Reports are systematically generated using predefined templates with predefined questions and answer options. Efforts to develop structured reporting have been initiated by radiology societies. Despite substantial backing from professional organizations such as the European Society of Radiology (ESR), it has not been widely adopted in Europe due to a lack of standardized reporting templates and insufficient incentives for its implementation [7,8].

The quality of radiology reports depends on how well the needs of referring physicians are met. However, the lack of direct communication and feedback between referring physicians and radiologists often creates gaps in understanding the clinical utility of these reports. Referring physicians might hesitate to express confusion, leading radiologists to overestimate the quality of their reports. Radiologists, especially

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radiology residents, often develop reporting skills passively by reading reports from others and emulating their approach, which perpetuates structural problems of radiology reports furthermore [9]. Case conferences address the issue of interdisciplinary communication by bringing specialists together to discuss complex cases.

This study primarily explored the preferences of referring physicians regarding radiology reports, aiming to identify key factors that enhance their clinical utility.

2. Methods

Ethical approval was not required, as the survey collected only physicians' opinions. The study adhered to ethical guidelines and all analyses were conducted in compliance with local data protection regulations.

2.1. Inclusion and exclusion criteria

The prospective survey study, conducted from June 2023 to March 2025, included licensed surgeons, internists, and general practitioners in rural and urban Germany who regularly request radiological services, excluding non-practicing physicians. Participants were randomly selected across all hierarchical levels. The exclusion criteria were retirement, not requesting radiology services and not filling up demographics or at least one survey question. (Fig. 1).

2.2. Survey design

A pilot test involving 10 physicians was conducted to refine the survey instrument, with their feedback leading to revisions of the questions. Following the pilot phase, randomly selected physicians were contacted via via email. A reminder sent after four weeks. To minimize order effects, the question order was randomized. Anonymity was maintained to reduce response bias.

Survey completion time was measured as the duration between the participant's survey start and finish. Completion times that were significantly above or below the interquartile range (IQR) or fell outside the typical range were considered outliers and excluded from the analysis.

2.3. Question formats

Slider Scale Questions

A slider scale question measured how frequently radiological findings fully addressed the referral question, using a satisfaction score from −100 (Never) to + 100 (Always), divided into 10-point intervals. Another slider (−50 to + 50) assessed preferences for concise (−50 to −20), medium (−10 to + 10), or detailed (+20 to + 50) reports. A third slider (−100 to + 100) evaluated reporting style preferences, with values indicating free-text (−100 to −50), mixed reporting (−50 to + 50), or structured reporting (+50 to + 100). (Supplementary) Mixed reporting combines free-text with structured reporting for clarity.

Single-choice questions.

Respondents selected the main reason why radiologists may not fully address the clinical question. They identified methods to enhance the readability and usability of radiology findings and the type of information often missing in reports. (Supplementary).

A 7-point Likert scale assessed how often referring physicians struggle to locate relevant details in radiology reports (1 = never, 7 = always). Referring physicians rated the helpfulness of multidisciplinary meetings (e.g., tumor boards) in understanding radiology reports using a 7-point Likert scale, with 1 = not helpful and 7 = very helpful. (Supplementary).

2.4. Statistical analysis

Descriptive statistics, including mean, median, mode and standard deviation were employed to summarize the sample. Statistical analyses included one-way ANOVA, chi-square tests for categorical variables, one-sample proportion z-tests and one-sample t-tests with a significance threshold set at $p < 0.05$.

3. Results

3.1. Participant characteristics

Out of the 300 physicians initially considered for the study, 42 physicians were excluded: 20 physicians did not utilize radiology

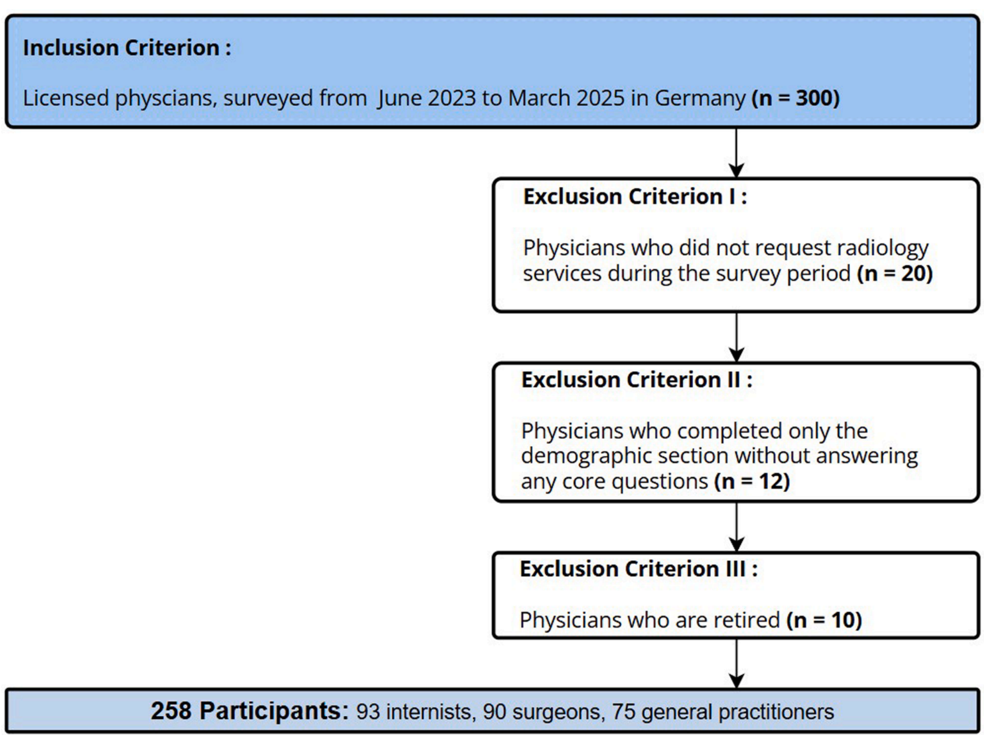

Fig. 1. Flow Chart of inclusion and exclusion criteria, standards of reporting The flow chart outlines the selection process for participants in the survey study.

Table 1
Specialties and experience of the participants.

	Number of participants	Average experience in years	Standarddeviation in years	Mode in years
Internists	120	9	6.4	9
Surgeons	110	10.5	11.24	10
General Practitioners	95	19.2	10.6	10

services, 12 completed only the demographic portion of the questionnaire without responding to core questions and 10 were retired. Ultimately, 258 participants were included: 93 internists, 90 surgeons and 75 general practitioners. (Fig. 1).

All 75 general practitioners worked in private practices, whereas internists and surgeons were employed in hospitals. Among 180 hospital doctors, 110 were assistant physicians, 79 were senior physicians and one doctor was a chief physician.

The average time to complete the survey, after trimming outliers, was 6:53 min. Internists had an average experience 9 years (median/mode: 9 years), surgeons 10.5 years (median: 11, mode: 10 years) and general practitioners 19.2 years (median: 18.5, mode: 10 years). (Table 1).

3.2. Satisfaction with radiology report completeness

The satisfaction score (38.4 ± 24.3) of the referring physicians indicated moderate satisfaction and was significantly above neutral ($p < 0.0001$). (Fig. 2) General practitioners were moderately satisfied (average 47.7 ± 21.3). Internists expressed similar satisfaction levels (internists: average 42.8 ± 22.9). Surgeons reported significantly the highest degree of dissatisfaction (average 15.3 ± 26), $p < 0.0001$.

3.3. Preferences for report length

55 % of the referring physicians preferred short, concise reports (−50 to −20 range). A smaller group (21.7 %) favored medium-length reports (−10 to + 10 range). In contrast, only 23.3 % of referring physicians preferred long, detailed reports (+10 to + 50 range). Referring physicians showed a significant preference for concise reports ($p < 0.001$). Surgeons demonstrated a significantly stronger preference for concise reports (mean: –23.9 ± 22.0) compared to internists (−12.37 ± 36.8, $p < 0.01$) and general practitioners (−14.6 ± 36.1, $p < 0.05$). (Fig. 3).

3.4. Preferences for free-text vs. structured reporting

25.7 % of referring physicians preferred free-text or unstructured reporting (−100 to −50), whereas 74.3 % favored either semi structured reporting (−50 to + 50) or fully structured reporting (+50 to + 100). (Fig. 4) Referring physicians showed a significant preference for structured reporting ($p < 0.001$).

Internists and surgeons showed significantly stronger preferences for structured reporting (internists: average 34.4 ± 56.9, surgeons: average 41.8 ± 56.3) compared to general practitioners (average −25.2 ± 56.9).

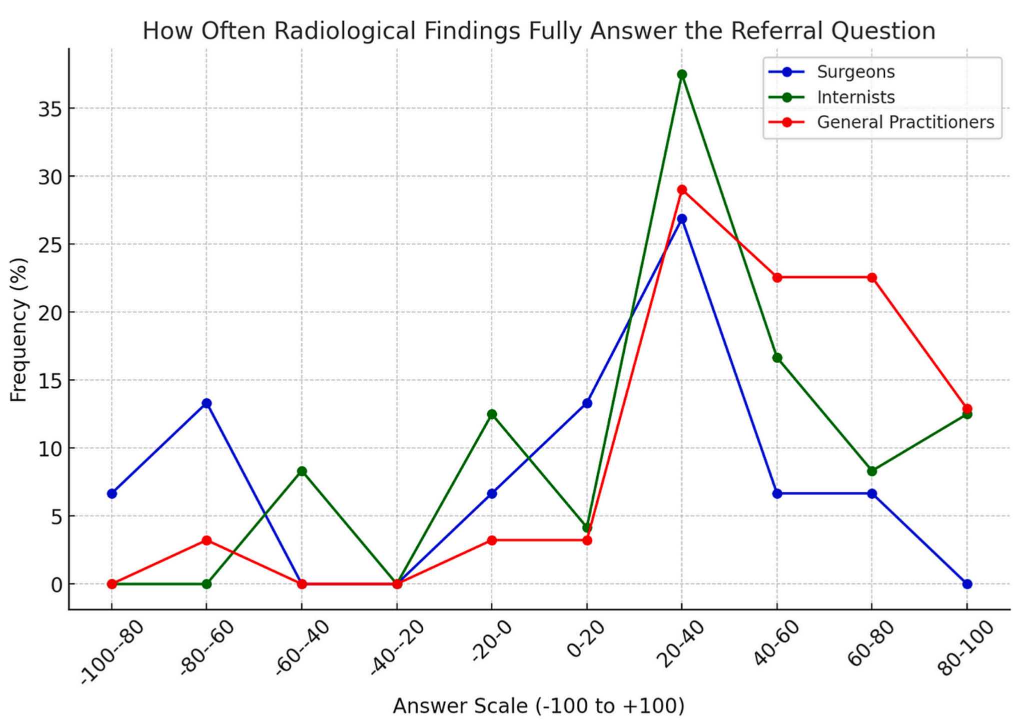

Fig. 2. Completeness of Radiology Reports of Referring Physicians Referring physicians rated completeness of radiology reports on a −100 to + 100 scale.

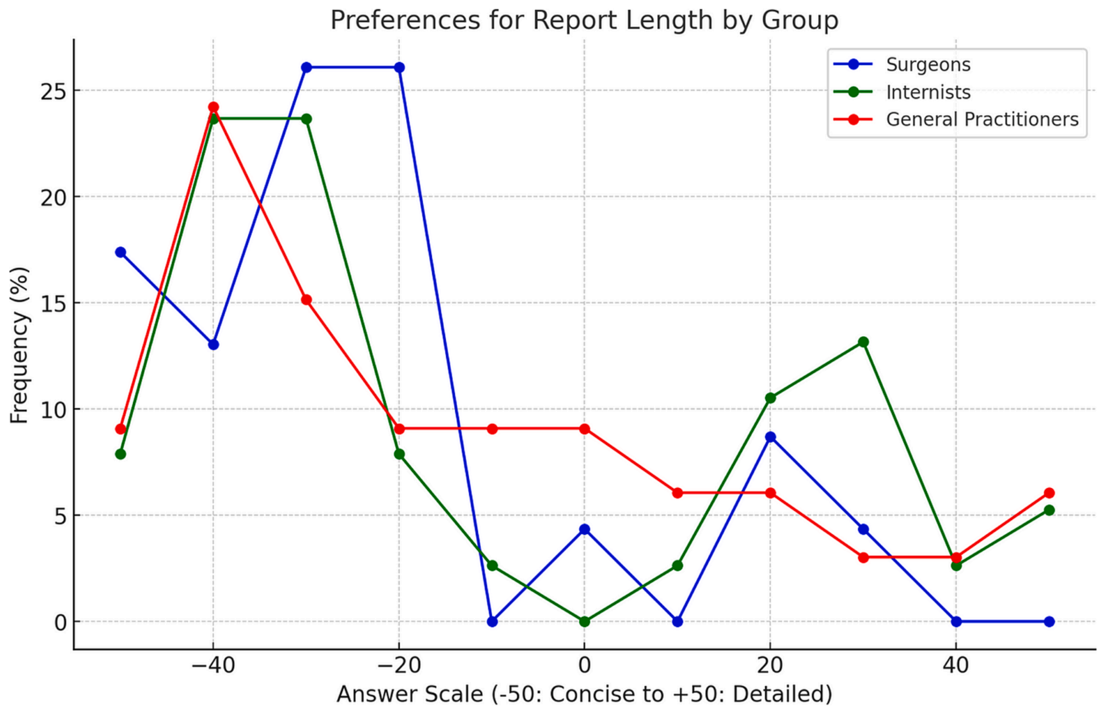

Fig. 3. Radiology Report Length Preferences of Referring Physicians The slider demonstrates preferences for concise (−50 to −20), medium (−10 to + 10), or detailed (+20 to + 50) reports.

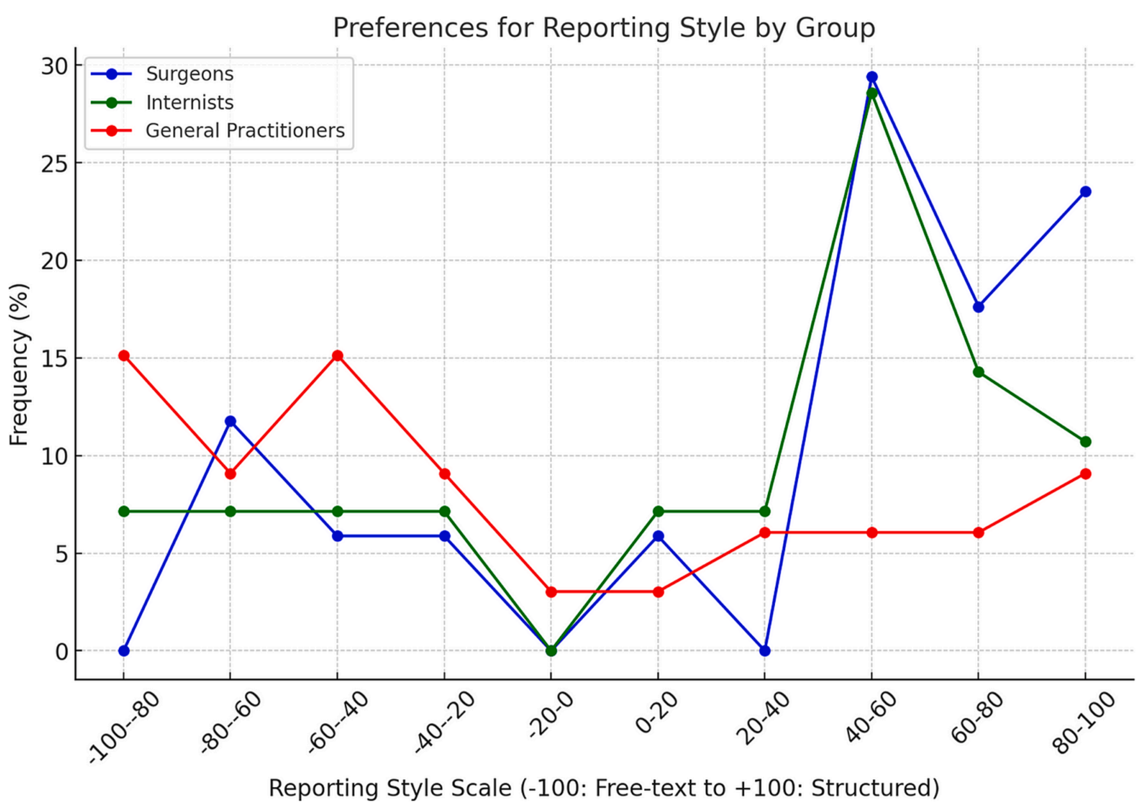

Fig. 4. Preferences for Free-Text vs. Structured Reporting of Referring Physicians The slider shows preferences for free-text reporting (−100 to −50), semi-structured reporting (−50 to + 50), and fully structured reporting (+50 to + 100).

3.5. Challenges in improving radiology reports

According to referring physicians, the main reasons for incomplete reports were “radiologists’ lack of understanding for the clinical context” (36.1 %), “missing preliminary examination imaging” (16 %), “inappropriate imaging techniques” (10.3 %), and “unclear or ambiguous question in the referral form” (12.6 %). Additional factors cited were “limited image quality” (9.2 %) and “missing communication”

(9.2 %), as well as “time pressure” (6.2 %). “Radiologists’ lack of understanding for the clinical context” was selected significantly more frequently than statistically expected under a normal distribution across all specialties (p < 0.001). Surgeons cited “unclear or ambiguous questions in the referral form” significantly more often than both general practitioners and internists (p < 0.05). In contrast, internists reported “limited image quality” significantly more frequently compared to both surgeons and general practitioners (p < 0.05). (Fig. 5).

3.6. Readability in radiology reports

Referring physicians reported few to moderate difficulties in finding medical information in radiology reports. The majority encountered moderate difficulty (63.2 %, 3–5 stars) in finding medical information, while 32.5 % experienced low difficulty (1–2 stars) and 4.3 % faced high difficulty (6–7 stars). Moderate difficulty (3–5 stars) was selected significantly more frequently than statistically expected under a normal distribution across all specialties (p < 0.01).

3.7. Approaches to improve readability

43.8 % of referring physicians favored “structured reporting” for improved readability and usability, while 26.5 % preferred a “concise impression section”. (Table 2) “Concise impression section” was significantly more frequently cited by general practitioners compared to both internists and surgeons. Conversely, “structured reporting” was cited significantly more frequently by surgeons and internists than by general practitioners (p < 0.05). (Fig. 6).

Table 2
How could the readability and clinical usability of radiological findings be increased?

Number of responses: 258	chosen	
Numbering of pathologies	36	13,91 %
Sensible paragraphs (logical structure)	41	15,89 %
Concise impression section	68	26,49 %
Structured reporting	113	43,7%

3.8. Missing information in radiology reports

Referring physicians selected ’determination of a diagnosis’ as the most frequently chosen option (39 %), followed by “recommendations for further diagnostic measures or follow-up examinations” (23 %). “Determination of a diagnosis” was cited significantly more often than statistically expected under a normal distribution across all specialties (p < 0.001). (Table 3).

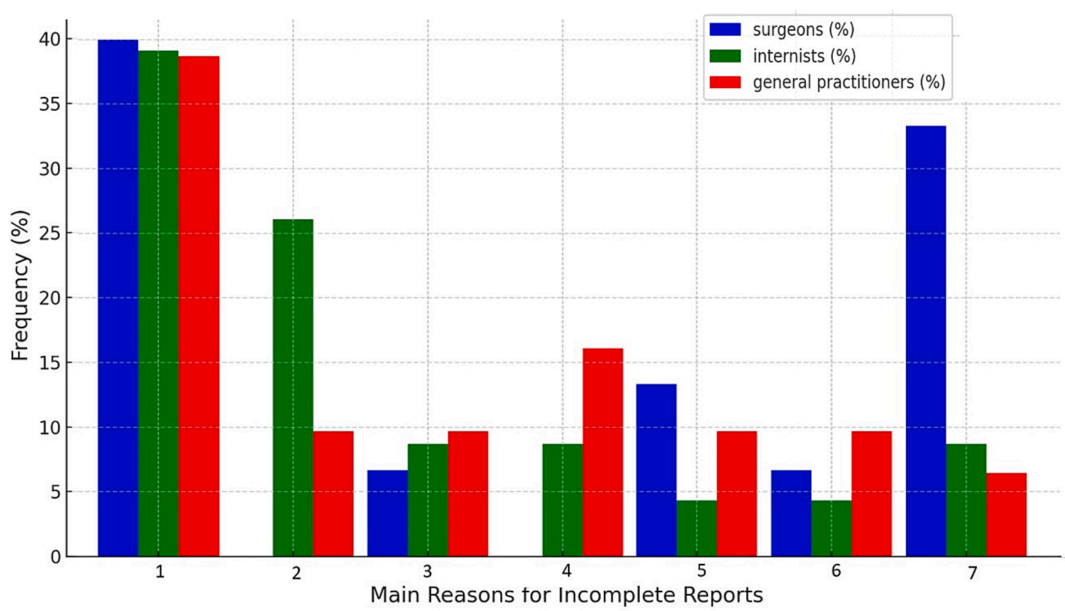

Fig. 5. Reasons for incomplete reports of Referring Physcians In a single-choice question, referring physicians identified the primary reason for incomplete report.

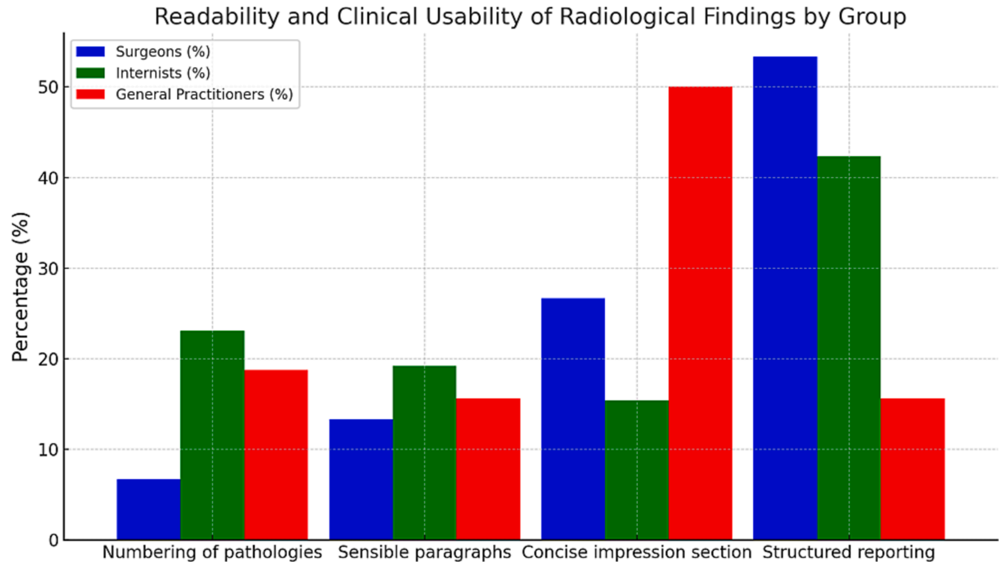

Fig. 6. Readability and Clinical Usability of Radiological Findings In a single-choice question, referring physicians identified methods to enhance the readability and usability of radiology findings.

Table 3
What type of information do you often find missing in radiology reports?

Number of responses: 258	chosen	
Measurements and sizes (e.g. of tumors or other pathologies)	29	11,24 %
Comparison with previous studies	41	15,97 %
Precise localization description of pathologies	27	10,65 %
Recommendations for further diagnostic measures or follow-up examinations	60	23,07 %
Determination of a diagnosis	101	39,07 %

3.9. *Importance of multidisciplinary meetings*

33.6 % of the referring physicians rated multidisciplinary meetings as very helpful in understanding radiology reports (7 stars), with an additional 51.3 % of referring physicians giving positive ratings of 5 or 6 stars. Only 1.4 % of referring physicians found them not helpful at all (1 star). The average rating was 5.7 ± 1.4 stars. All specialities rated the value of multidisciplinary meetings for report understanding significantly above neutral (4 stars) ($p < 0.001$). Internists perceived the helpfulness of multidisciplinary meetings in understanding radiology reports significantly more positively compared to surgeons and general practitioners ($p < 0.05$). (Fig. 7).

4. Discussion

Our analysis reveals several main findings that highlight opportunities for improving the quality of radiology reports in regards to referring physicians.

First, concise and structured reporting is preferred by surgeons and internists whereas geneneral practitioners favor free-text reporting. Second, referring physicians in the survey identified “diagnosis determination” and “recommendations for further diagnostics or follow-up examinations” as commonly missing in radiology reports. Third, referring physicians in the survey most frequently selected “radiologists’ lack of understanding of the clinical context” as the main reason for incomplete radiology reports. Lastly, referring physicians strongly support

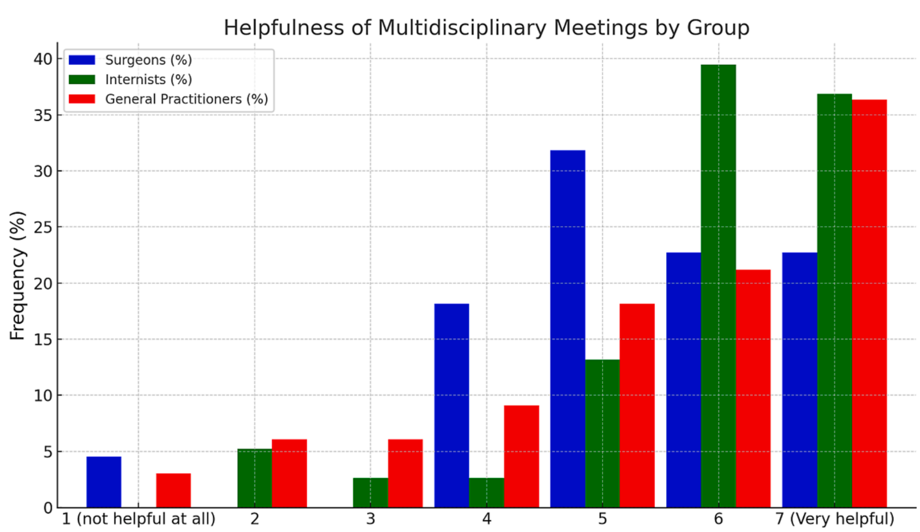

Fig. 7. The Role of Multidisciplinary Meetings in Enhancing Report Comprehension of Referring Physicians Referring physicians rated the usefulness of multidisciplinary meetings in understanding radiology reports on a 7-point scale, where 1 = not helpful at all and 7 = very helpful.

multidisciplinary meetings to improve the understanding of radiology reports.

Preferences for free-text reporting versus structured reporting varied in our study. Reasons for that may be found in the different ways these medical specialties interact with radiological findings. General practitioners may prefer free-text reporting for its flexibility in addressing a wide range of conditions. Internists and surgeons may favor structured reporting for precise, standardized responses to specific clinical queries. Hybrid formats combining structured elements with optional free-text sections could meet the needs of both specialties.

Our findings on the preference for structured reporting among surgeons and internists match those of Stanzione et al. (2021) who showed that structured reporting in chest CT for COVID-19 patients improves referring physician satisfaction. [10] Additionally, a review by Stanzione demonstrated that it enhances consistency in abdominal MRI [11]. Furthermore, a study by Hawkins et al. showed that introducing structured reporting templates reduced errors caused by speech recognition software, including confusing phrases or term misinterpretations [12].

The American College of Radiology (ACR) has pioneered standardized reporting systems such as BI-RADS and LI-RADS in the United States of America, which have been adopted nationwide ever since. These systems standardize breast and liver imaging reporting, improving communication and reports [13–16].

The widespread adoption of structured reporting in Europe has been slower. Differences in adoption rates between European countries, as identified by dos Santos et al., may be attributed to cultural and organizational factors, such as varying healthcare systems, resources, and professional practices [8]. Furthermore, structured reporting requires radiologists to shift focus between images and the reporting template, disrupting workflow [8,17]. Managing complex cases often becomes more time-consuming with template-based formats. Incidental findings in those cases do not fit within structured reporting templates because these templates are typically designed to address specific clinical questions and suspected diagnoses [18]. In a broader context, the differing preferences of referring physicians—who are the primary audience of radiology reports—could present a challenge to the widespread adoption of structured reporting systems.

We demonstrated that most referring physicians prefer concise reports. Research indicates that referring physicians frequently overlook the findings section, emphasizing the importance of the impression section [19]. We believe that an increasing number of concise reports might make it less likely for referring physicians to skip the findings section. Radiologists must balance conciseness with the need for complete documentation to minimize medico-legal risks, such as the omission of relevant findings [20,21].

In our survey study, most referring physicians indicated that recommendations for further diagnostic measures or follow-up exams and definitive diagnoses are often missing in radiology reports. Follow-up imaging recommendations for abnormal findings appear in 2–5 % of radiology reports [22].

For instance, benign lesions should be explicitly classified and clearly indicated as not requiring further diagnostic procedures. Stating “This benign lesion does not warrant additional evaluation” in the report may help minimize unnecessary investigations. If this assessment or recommendation is missing, the referring physician must either make their own classification or contact the radiologist [23].

The issue of recommending follow-up imaging is clinically complex. Some radiologists hesitate to provide recommendations, as they may lack the full clinical picture. For example, recommending a follow-up CT for pneumonia may be unnecessary if the patient is already improving on antibiotic therapy, a detail not visible on imaging. Furthermore, the establishment of a definitive diagnosis in radiology reports provides referring physicians with clear guidance and certainty in clinical decision-making. This poses a challenge for radiologists, as imaging findings are often difficult to classify definitively. In clinical practice, definitive diagnoses are the exception and radiologists typically use terms like “consistent with” or “suggestive of” to indicate diagnostic certainty. “Consistent with” signals higher confidence in the diagnosis, while “suggestive of” implies support for a diagnosis but lacks sufficient specificity.

Khorasani et al. found significant discrepancies between radiologists and non-radiologists in interpreting diagnostic certainty from commonly used phrases in radiology reports. This underscores the importance of standardizing diagnostic certainty terms to ensure clear and consistent communication [24]. Targeted training for radiologists can encourage the use of more precise language, allowing radiologists to convey their findings with increased confidence and clarity, while reducing overly cautious phrasing [25].

In this study, all specialties identified a lack of understanding of the clinical context of radiologists as the main cause of incomplete reports.

Without sufficient background information, radiologists may find it difficult to provide proper context in their reports. Previous studies have shown that including clinical information in radiology reports enhances interpretation accuracy [26,27]. More generally, limited pre-imaging consultation between referrers and radiologists, the exclusion of radiologists from assessing imaging appropriateness at the time of the imaging request and the perception of imaging as a fait accompli on the day of the imaging examination contribute to incomplete reports in regards to all participants [28–30].

Strengthening interdisciplinary collaboration in all aspects is therefore essential to improving the clinical relevance, accuracy and impact of radiology reports. Our findings reinforce the value of multidisciplinary meetings in understanding radiology reports for referring physicians. Research by Dendl et al. showed that such discussions resulted in diagnostic or therapeutic changes in 37 % of cases, highlighting their significance in enhancing patient care [31]. These meetings refine communication between radiologists and referring physicians, reducing medical errors and unnecessary imaging, which in turn lowers associated treatment costs. For example, structured bilateral feedback between radiologists and ICU physicians has been shown to prevent imaging-related medical errors by re-evaluating recent examinations, allowing for early detection and correction. This proactive approach enhances patient safety and optimizes resource utilization [32]. However, the opportunity costs of these meetings are significant. A study at a Midwest Academic Medical Center found that radiologists spent 3.358 h annually in these meetings – equivalent to two full-time positions – resulting in $1.2 million in unreimbursed costs [33]. Despite these financial considerations, the clinical advantages outweigh the disadvantages. To balance clinical benefits with economic sustainability institutions should explore targeted case discussions and structured meeting formats.

Several limitations must be discussed. First, this study relied on self-reported data, which can introduce response bias, as participants may provide socially desirable answers or incomplete information. In our study, we employed strategies to mitigate response bias, such as anonymizing responses and using random sampling. Second, the survey format may have constrained responses by offering predefined options. Third, the study had a large proportion of assistant physicians, who may have different experiences or expectations, compared to senior physicians. Fourth, the questions concerning structured reporting were not context-based, as it did not account for the specific clinical scenarios or conditions. Fifth, while the study explored various aspects of radiology reports, such as report length, structure, and completeness, it did not specifically analyze how these factors influence referring physicians' decisions or clinical outcomes. Finally, the study focused on internists, surgeons, general practitioners, potentially overlooking the perspectives of other specialties. Further studies should include subgroup analysis to assess the impact of structured reporting on specific types of reports, such as those for oncological or musculoskeletal conditions, and evaluate whether structured formats offer distinct advantages in these contexts.

In conclusion, this study emphasizes the importance of radiologists aligning their reports with the needs of referring physicians and improve interdisciplinary communication to ensure more accurate and understandable radiology reports.

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P.R. led the research design, data collection, analysis, and manuscript preparation. All authors contributed to data interpretation and assisted in manuscript review and editing.

Author contributions and disclosures: All authors have contributed significantly to this work, participating in the conception, design, analysis, interpretation of data, and writing/editing of this manuscript. P.R. wrote the primary draft of the manuscript and designed the research. The other authors were involved in the analysis of the data and performed research or assisted with the technical development.

CRediT authorship contribution statement

Philipp Reschke: Writing – original draft, Project administration, Data curation, Conceptualization. **Christian Booz:** Writing – review & editing, Supervision. **Leon D. Gruenewald:** Writing – review & editing. **Vitali Koch:** Writing – review & editing. **Elena Höhne:** Writing – review & editing. **Aynur Gökduman:** Writing – review & editing, Resources. **Katrin Eichler:** Writing – review & editing, Formal analysis. **Jörg Schlüchtermann:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Thomas J. Vogl:** Writing – review & editing, Validation. **Jennifer Gotta:** Writing – review & editing, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejrad.2025.112068.

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